

Chapter 3: THE RELATIONAL DATA MODEL **&** **RELATIONAL DATABASE CONSTRAINTS**

Figure: One possible database state for the COMPANY relational database schema.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

1. Suppose each of the following Update operations is applied directly to the database of above Figure, Discuss *all* integrity constraints violated by each operation, if any, and the different ways of enforcing these constraints:

a. Insert <'Robert', 'F', 'Scott', '943775543', '1972-06-21', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555', 1> into EMPLOYEE.

Violation	Entity integrity	Key constraint	Referential integrity
No	-	-	-

b. Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT.

Violation	Entity integrity	Key constraint	Referential integrity
Yes	-	-	DNUM:2 -No tuple in the DEPARTMENT relation with DNUMBER=2 exists. Ways of enforcing: -Changing the DNUM to either 5, 4, 1 -Inserting a new tuple of DNUM=2 in the DEPARTMENT itself.

c. Insert <'Production', 4, '943775543', '2007-10-01'> into DEPARTMENT.

Violation	Entity integrity	Key constraint	Referential integrity
Yes	-	DNUMBER=4 -It already exists in the DEPARTMENT tuple. Ways of enforcing: -Changing the value of DNUMBER which doesn't exist and also not violate the key constraint.	MGRSSN='943775543' - No tuple exists in the EMPLOYEE relation with SSN='943775543'. Ways of enforcing: -Changing the value of MGRSSN to an existing SSN value in EMPLOYEE. -Inserting a new EMPLOYEE tuple with SSN='943775543'.

d. Insert <'677678989', NULL, '40.0'> into WORKS_ON.

Violation	Entity integrity	Key constraint	Referential integrity
Yes	PNO=NULL -It cannot be NULL as it is the primary key in WORKS_ON Ways of enforcing:	-	ESSN='677678989' -No tuple exists in the EMPLOYEE relation with SSN='677678989'.

	-Changing the value of PNO and also the value of PNUMBER should exist in the PROJECT relation.		Ways of enforcing: -Changing the value of ESSN to an existing SSN value in EMPLOYEE. -Inserting a new EMPLOYEE tuple with SSN='677678989'.
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e. Insert <'453453453', 'John', 'M', '1990-12-12', 'spouse'> into DEPENDENT.

Violation	Entity integrity	Key constraint	Referential integrity
No	-	-	-

f. Delete the WORKS_ON tuples with Essn = '333445555'.

Violation	Entity integrity	Key constraint	Referential integrity
No	-	-	-

g. Delete the EMPLOYEE tuple with Ssn = '987654321'.

Violation	Entity integrity	Key constraint	Referential integrity
Yes	-	-	Ssn = '987654321' -Many tuples exist in the WORKS_ON, DEPENDENT, DEPARTMENT, and EMPLOYEE relations that reference the tuple being deleted from EMPLOYEE. Ways of enforcing: -Deleting all tuples in the WORKS_ON, DEPENDENT, DEPARTMENT, and EMPLOYEE relations whose values for ESSN, ESSN, MGRSSN, and SUPERSSN.

h. Delete the PROJECT tuple with Pname = 'ProductX'.

Violation	Entity integrity	Key constraint	Referential integrity
Yes	-	-	Pname = 'ProductX' -Affects the other tuple which exists in the WORKS_ON relates that reference the tuple being deleted from PROJECT. Ways of enforcing: -Deleting the tuples in the WORKS_ON relation whose value for PNO=1

- i. Modify the Mgr_ssn and Mgr_start_date of the DEPARTMENT tuple with Dnumber = 5 to '123456789' and '2007-10-01', respectively.

Violation	Entity integrity	Key constraint	Referential integrity
No	-	-	-

- j. Modify the Super_ssn attribute of the EMPLOYEE tuple with Ssn = '999887777' to '943775543'.

Violation	Entity integrity	Key constraint	Referential integrity
Yes	-	-	SUPERSSN='943775543' -No tuple exists in the EMPLOYEE relation with SSN='943775543'. Ways of enforcing: -Inserting a new EMPLOYEE tuple with SSN='943775543'.

- k. Modify the Hours attribute of the WORKS_ON tuple with Essn = '999887777' and Pno = 10 to '5.0'.

Violation	Entity integrity	Key constraint	Referential integrity
No	-	-	-

2. Consider the relation CLASS (Course#, Univ_Section#, Instructor_name, Semester, Building_code, Room#, Time_period, Weekdays, Credit_hours). This represents classes taught in a university, with unique Univ_section#. Identify what you think should be various candidate keys, and write in your own words the conditions or assumptions under which each candidate key would be valid.

Any relation schema may have more than one key. In this case, each of the keys is called a candidate key. It is common to designate one of the candidate keys as the primary key of the relation. This is the candidate key whose values are used to identify tuples in the relation. All possible candidate keys include the following:

1. **(Univ_Section#)** - If it is unique across all semesters.
2. **(Semester, BuildingCode, Room#, TimePeriod, Weekdays)** - If more than one course cannot be conducted in the same room during a particular semester.
3. **(InstructorName, Semester)** - If an instructor can only teach one course in each semester.
4. **(Course#, Univ_Section#, Semester)** - would be the candidate key, If **(Univ_Section#)** is not unique across all semesters.